More dramatic was the testing of an igniter to be used for the Centaure sounding rocket for a sodium vapour experiment. The year was 1967. The French rocket was designed to fly to an altitude of about 100 km where the payload would disperse sodium vapour in the atmosphere. This would be photographed from several locations around the globe. My job was to make the igniter work. This was to be done by using a squib with a small measure of an explosive at the tip, which would in turn ignite gunpowder (G20) to open the payload and send the sodium vapour out.

Four rockets were to be launched from Thumba, at an interval of four hours. Ground observation stations across the globe were kept ready and the world was looking eagerly at what this space toddler called India could do. Murthy was confident of our success, so was I—until I walked into the office library the day before the launch, and stumbled upon a 1942 paper by scientists Able and Noble.

It had a simple point that threatened to defuse our Centaure launch: gunpowder, specifically G20, behaves like sand while the pressure is less than one-fourth of that on ground. It meant that gunpowder just wouldn't fire at an altitude of 100 km. It meant that our igniter would not work. It meant that sodium vapour would not be ignited to discharge its vapour in the atmosphere.

I went to Kalam. He dismissed it initially as some cock-and-bull study. But when I insisted that the study sounded serious, he said we would test it on ground. Again, we had no simulator or a laboratory where upper atmosphere-like conditions can be created. Necessity may be the father of invention; here desperation was the mother of all adventurous innovations. I got a bell jar and kept some gunpowder in touch with a squib. Y S Rajan, an astrophysicist from Physical Research Laboratory who later became the scientific secretary of ISRO and a conscience keeper of Satish Dhawan, offered a vacuum pump. We connected the pump to the sealed bell jar to bring down the pressure to one-fourth. Now we had to see if the gunpowder ignited. It did not. Able and Noble were right.

I repeated the experiment a dozen times, and a dozen times it failed—or succeeded, depending on the way you looked at it. Kalam wanted to see the